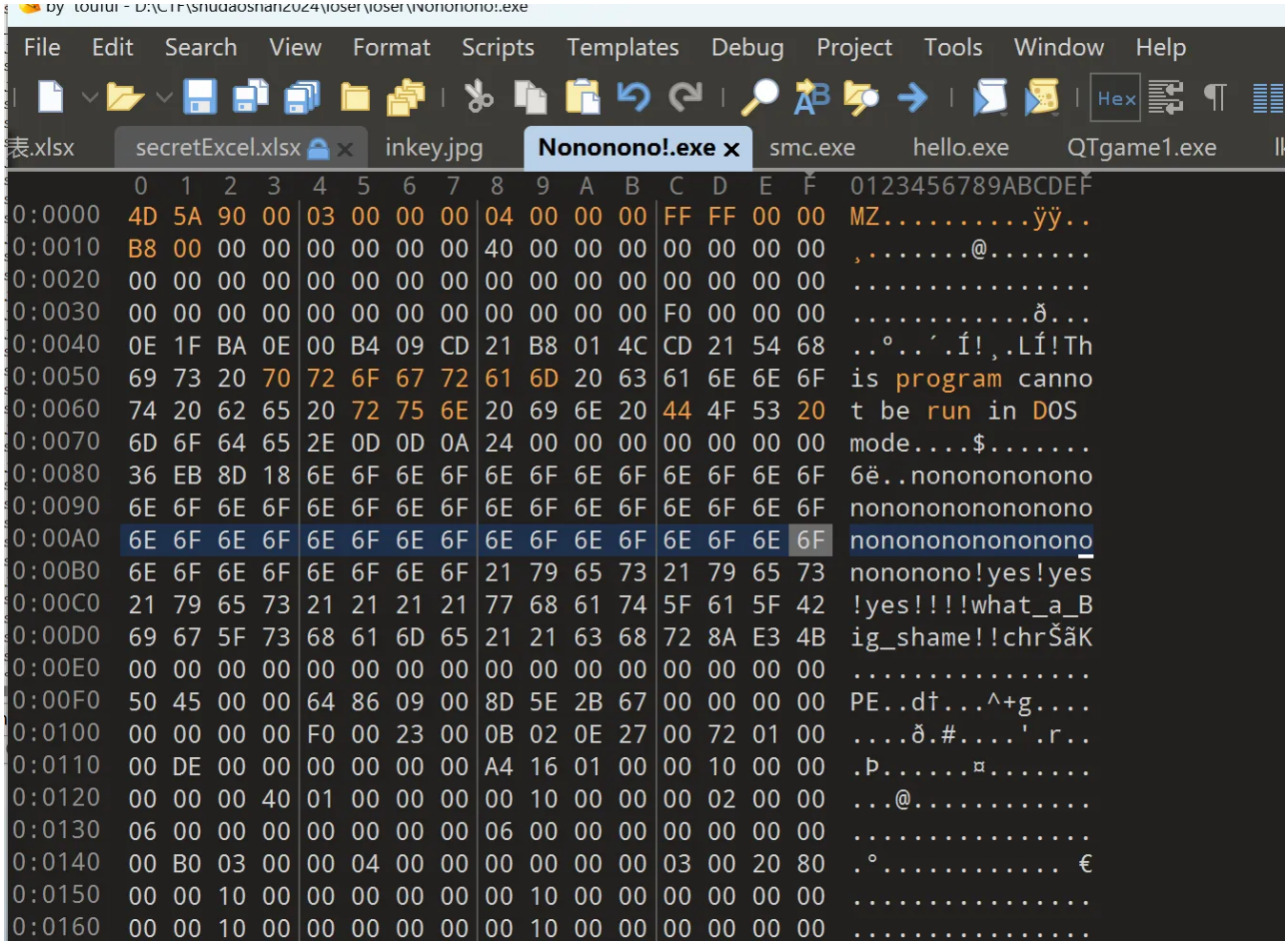
Reverse

非常难忘的赤石体验

最高的山最长的河

简单地修一下文件头，唉，又被大飞老师折磨疯了一位



cpp逆向，但是根本难不倒他，无非就是一些数据类型的转换，最后看出就是base64魔改+tea

tea没什么好说的，就是程序里跑的是解密脚本

加密流程如下

```
#include<iostream>
#include<stdint.h>
using namespace std;

unsigned char
input[]="ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuv";
unsigned char table[] =
"ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/";
unsigned char enc;

void decrypt (uint32_t* v, uint32_t* k) {
    uint32_t v0=v[0], v1=v[1], i;
    uint32_t delta=0x61C88647;
    uint32_t sum = 0xC6EF3720;
    uint32_t k0=k[0], k1=k[1], k2=k[2], k3=k[3];
    for (i=0; i<32; i++) {                          //解密时将加密算法的
顺序倒过来，还有+=变为-=
        v1 -= ((v0<<4) + k2) ^ (v0 + sum) ^ ((v0>>5) + k3);
        v0 -= ((v1<<4) + k0) ^ (v1 + sum) ^ ((v1>>5) + k1);
        sum += delta;
        //printf("%x %x \n", v0, v1);
    }
    v[0]=v0 ^ k[4]; v[1]=v1 ^ k[4];//解密后再重新赋值
    //printf("\n");
}

//MTaVE7TVo3vVfvoV3ofVvf3Vo3vVfvoV3vfV3vfV3vfV39hR

int base64(unsigned char *input, unsigned char*output){
    unsigned char a, b , c, d, e, f;
    int j = 0;
    for(int i = 0; i < 36; i+=3) {
        a = input[i] & 0x3F;
        b = input[i + 1] & 0x3F;
        c = input[i + 2] & 0x3F;

        d = 16 * ((input[i] >> 6) & 3);
```

```
        e = d + 4 * ((input[i + 1] >> 6) & 3);
        f = ((input[i + 2] >> 6) & 3) + e;
        output[j++] = table[a];
        output[j++] = table[c];
        output[j++] = table[b];
        output[j++] = table[f];
    }

}

int main()
{
    unsigned char ci[100] = {0};


    base64(input, ci);


    uint32_t k[]={0x4C, 0x5A, 0x53, 0x44, 0x53};

    for(int i = 0; i < 12; i+=2) {
        uint32_t *v = (uint32_t*)ci + i ;
        decrypt(v, k);
        //printf("%x %X \n", v[0], v[1]);
    }

    for(int i = 0; i < 48; i++) {
        printf("0x%02X, ", ci[i]);
    }

}
```

tea解密脚本

```
#include<iostream>
#include<stdint.h>
using namespace std;
```

```
unsigned char
input[]="ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuv";
unsigned char table[] =
"ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/";
unsigned char cipher[] = {0x04, 0x9f, 0x64, 0xa5, 0x17, 0xa6, 0xb7,
0x39, 0x4d, 0x26, 0xcd, 0x22, 0xdb, 0x82, 0x10, 0x7d, 0x39, 0x27,
0xdb, 0x85, 0xa5, 0x1e, 0x3c, 0x42, 0x4d, 0x26, 0xcd, 0x22, 0xdb,
0x82, 0x10, 0x7d, 0x80, 0x68, 0x37, 0xd6, 0x19, 0x15, 0xf9, 0xb1,
0x24, 0x42, 0x18, 0x44, 0x2c, 0xa2, 0x78, 0x8f};

void encrypt (uint32_t* v, uint32_t* k) {
    uint32_t v0=v[0], v1=v[1], sum=0, i;     //v0,v1分别为字符串的低字
节高字节
    uint32_t delta=0x61C88647;
    uint32_t k0=k[0], k1=k[1], k2=k[2], k3=k[3];

    v0 ^= k[4];
    v1 ^= k[4];
    for (i=0; i < 32; i++) {                //加密32轮
        sum -= delta;
        //printf("%x %x \n", v0, v1);
        v0 += ((v1<<4) + k0) ^ (v1 + sum) ^ ((v1>>5) + k1);
        v1 += ((v0<<4) + k2) ^ (v0 + sum) ^ ((v0>>5) + k3);

    }
    v[0]=v0; v[1]=v1;//加密后再重新赋值
}

int main()
{
    unsigned char a, b , c, d, e, f;


    uint32_t k[]={0x4C, 0x5A, 0x53, 0x44, 0x53};

    for(int i = 0; i < 12; i+=2) {
        uint32_t *v = (uint32_t*)cipher + i;
        encrypt(v, k);
```

```
        //printf("%x %X \n", v[0], v[1]);
    }

    for(int i = 0; i < 48; i++) {
        printf("%c", cipher[i]);
    }
}
```

众所周知，base64类编码不需要技巧，只需要打表就好了，对照一下拿flag

```
#include<iostream>
#include<stdint.h>
using namespace std;

unsigned char
input[]="ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuv";
unsigned char table[] =
"ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/";
unsigned char
enc[]="MTaVE7TVo3vVfvoV3ofVvf3Vo3vVfvoV3vfV3vfV3vfV39hR";

//MTaV  E7TV    o3vV    fvoV    3ofV    vf3V    o3vV    fvoV
3vfV    3vfV    3vfV    39hR
//LZSDS{how_how_how_how_how_ow_ow_ow!}


int base64(unsigned char *input, unsigned char*output){
    unsigned char a, b , c, d, e, f;
    int j = 0;
    for(int i = 0; i < 3; i+=3) {
        a = input[i] & 0x3F;
        b = input[i + 1] & 0x3F;
        c = input[i + 2] & 0x3F;

        d = 16 * ((input[i] >> 6) & 3);
        e = d + 4 * ((input[i + 1] >> 6) & 3);
        f = ((input[i + 2] >> 6) & 3) + e;
        output[j++] = table[a];
        output[j++] = table[c];
```

```
        output[j++] = table[b];
        output[j++] = table[f];
    }

}

int main()
{
    freopen("table.txt", "w", stdout);
    unsigned char input[4] = {0, 0, 0, 0};
    unsigned char output[5] = {0};
    for(int i = 33; i <= 127; i++) {
        for(int k = 33; k <= 127; k++) {
            for(int j = 33; j <= 127; j++) {
                input[0] = i;
                input[1] = k;
                input[2] = j;
                base64(input, output);
                for(int l = 0; l < 3; l++) {
                    printf("%c", input[l]);

                }
                printf("  ");
                for(int l = 0; l < 4; l++) {
                    printf("%c", output[l]);

                }
                printf("\n");
            }
        }
    }

}
```

Hello Harmony

非常简单的鸿蒙逆向，甚至还给了x86架构的

```
#include<iostream>
using namespace std;

unsigned char key[] = "HelloSDS";
unsigned char box[257];
int visit[256];
unsigned int key2[9];

unsigned char output[] = {  0xF6, 0xB0, 0xA6, 0x36, 0x9A, 0xB3,
0x2B, 0xBF, 0x94, 0x54,
  0x15, 0x97, 0x93, 0x59, 0xBF, 0x50, 0x4D, 0xBF, 0x0A, 0x59,
  0x06, 0xD7, 0x97, 0x50, 0xD6, 0x59, 0x54, 0xD7, 0xCF, 0x06,
  0x5D, 0x20, 0x1D, 0x5A, 0x22, 0xEE, 0x99, 0x1F, 0xE1, 0x18};

unsigned char search(unsigned char a) {
    int counts = 0;
    unsigned char res = 0;
    for(int i = 0; i < 256; i++) {
        if(box[i] == a){
            res = i;
        }
    }
    return res;
}

int main()
{
    for(int i = 0; i < 256; i++) {
        box[i] = (0xA7 * i + 0xAD) % 0x100;
        //printf("%x ", box[i]);
    }

    int  len = 8;
    for(int i = 0; i < 8; i++) {
```

```
        key2[i] = (unsigned int)key[(i + 3) % len] | (unsigned int)
(key[(i + 2) % len] << 8) | (unsigned int)(key[(i + 1) % len] <<
16) | (unsigned int)(key[i % len] << 24);
    }
    unsigned char tmp;

    for(int i = 7; i >= 0; i--)
    {



        tmp = output[39];
        for(int k = 39; k >= 1; k--) {
            output[k] = output[k - 1];
        }
        output[0] = tmp;


        for(int k = 0;k < 40; k++) {
        output[k] = search(output[k]);
        }

        *(unsigned int*)output ^= key2[i];
        *((unsigned int*)output + 1) ^= key2[i];

        for(int k = 0; k < 40; k++) {
    }



    }

    for(int k = 0;k < 40; k++) {
        printf("%c", output[k]);


        }

}
```

最后大写ROT21，小写ROT19就出了

Super panda girl

用在线网站和平常用的脚本都解不出来，后来灵机一动直接复制到C#里运行就可以了

```
using System;
using System.Text;

// Token: 0x02000004 RID: 4
public class s5Ss55s5s5s5sSs5
{
    // Token: 0x06000009 RID: 9 RVA: 0x00003E08 File Offset:
0x00002008
    public bool ssSSs5s5s5(string SSS55sSS)
    {
        byte[] s555s5s = this.ssS5sss5s(SSS55sSS, "LZSDS");
        for(int i = 0; i < s555s5s.Length;i++)
        {
            Console.Write(Convert.ToChar(s555s5s[i]));
        }
        return this.CompareArrays(s555s5s, this.sssss55s);
    }

    // Token: 0x0600000A RID: 10 RVA: 0x00003E30 File Offset:
0x00002030
    private byte[] ssS5sss5s(string sSs5s5s5, string S5SSS5s5)
    {
        int length = S5SSS5s5.Length;

        //byte[] bytes = Encoding.UTF8.GetBytes(sSs5s5s5);
        byte[] bytes = {57,
        244,
        117,
        200,
        213,
        87,
        194,
        195,
```

```
        164,
        100,
        103,
        63,
        19,
        79,
        137,
        70,
        201,
        24,
        163,
        129,
        237,
        210,
        5,
        19,
        35,
        21};
        int length2 = bytes.Length;
        byte[] array = new byte[256];
        byte[] array2 = new byte[length2];
        for (int i = 0; i < 256; i++)
        {
            array[i] = (byte)i;
        }
        int num = 0;
        for (int j = 0; j < 256; j++)
        {
            num = (num + (int)array[j] + (int)S5SSS5s5[j % length])
% 256;
            byte b = array[j];
            array[j] = array[num];
            array[num] = b;
        }
        int num2 = 0;
        int num3 = 0;
        for (int k = 0; k < length2; k++)
        {
            num2 = (num2 + 1) % 256;
```

```
            num3 = (num3 + (int)array[num2]) % 256;
            byte b2 = array[num2];
            array[num2] = array[num3];
            array[num3] = b2;
            byte b3 = (byte)((int)(array[num2] + array[num3]) %
256);
            array2[k] = ((byte)(bytes[k] ^ b3));
        }
        return array2;
    }

    // Token: 0x0600000B RID: 11 RVA: 0x00003F30 File Offset:
0x00002130
    private bool CompareArrays(byte[] s555s5s, byte[] s5sSss5s)
    {
        if (s555s5s.Length != s5sSss5s.Length)
        {
            return false;
        }
        for (int i = 0; i < s555s5s.Length; i++)
        {
            if (s555s5s[i] != s5sSss5s[i])
            {
                return false;
            }
        }
        return true;
    }

    // Token: 0x04000007 RID: 7
    private const string S5SSS5s5 = "LZSDS";

    // Token: 0x04000008 RID: 8
    private readonly byte[] sssss55s = new byte[]
    {
        57,
        244,
        117,
        200,
```

```
        213,
        87,
        194,
        195,
        164,
        100,
        103,
        63,
        19,
        79,
        137,
        70,
        201,
        24,
        163,
        129,
        237,
        210,
        5,
        19,
        35,
        21
    };
}


class Program
{
    static void Main()
    {
        Console.WriteLine("Hello, World!");
        s5Ss55s5s5s5sSs5 a = new s5Ss55s5s5s5sSs5();
        bool b;
        b = a.ssSSs5s5s5("114514");
        Console.WriteLine(b);
    }
}
```

ezPotato

前段时间刚好在研究QTcpp，这下真的比家还要亲切，很快就找到了关键的函数，加密也是非常地简单

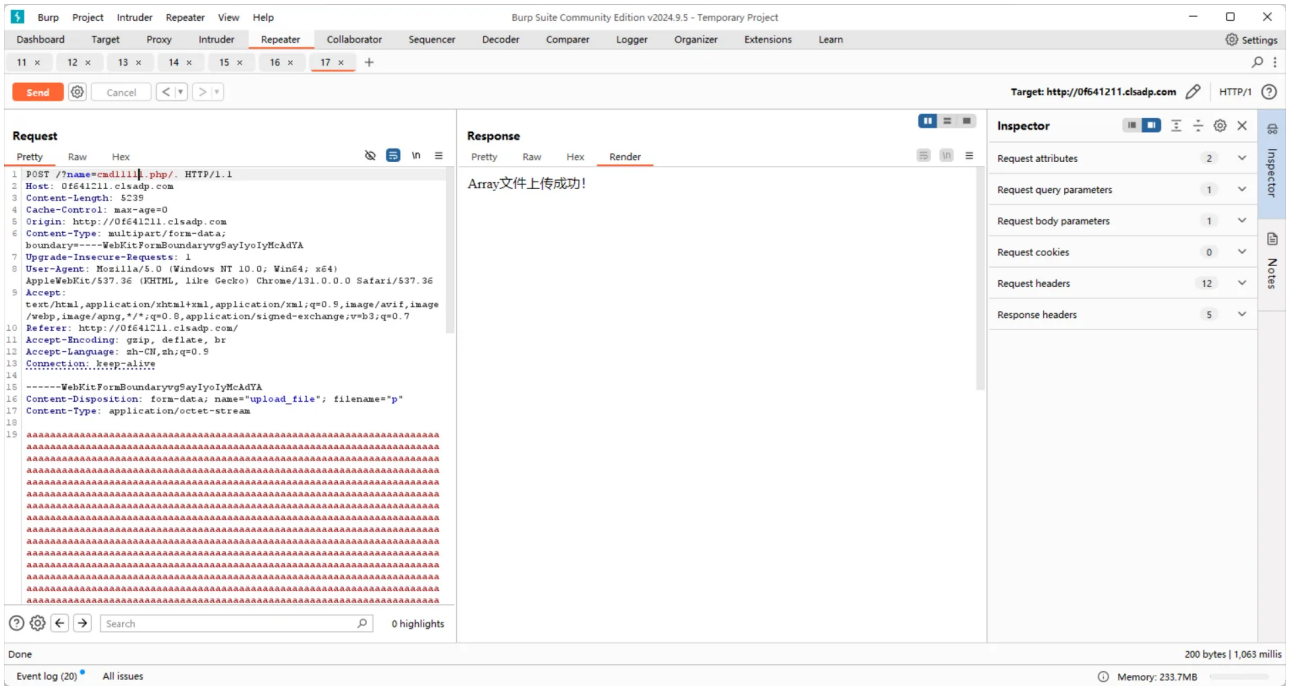
```
#include<iostream>
using namespace std;
unsigned char c[] =
{
  0x7D, 0x00, 0x00, 0x00, 0x2D, 0x00, 0x00, 0x00, 0x36, 0x00,
  0x00, 0x00, 0x37, 0x00, 0x00, 0x00, 0x32, 0x00, 0x00, 0x00,
  0x49, 0x00, 0x00, 0x00, 0x73, 0x00, 0x00, 0x00, 0x6B, 0x00,
  0x00, 0x00, 0x47, 0x00, 0x00, 0x00, 0x44, 0x00, 0x00, 0x00,
  0x17, 0x00, 0x00, 0x00, 0x0A, 0x00, 0x00, 0x00, 0x3E, 0x00,
  0x00, 0x00, 0x44, 0x00, 0x00, 0x00, 0x56, 0x00, 0x00, 0x00,
  0x46, 0x00, 0x00, 0x00, 0x48, 0x00, 0x00, 0x00, 0x28, 0x00,
  0x00, 0x00, 0x24, 0x00, 0x00, 0x00, 0x11, 0x00, 0x00, 0x00,
  0x52, 0x00, 0x00, 0x00, 0x46, 0x00, 0x00, 0x00, 0x41, 0x00,
  0x00, 0x00, 0x74, 0x00, 0x00, 0x00, 0x52, 0x00, 0x00, 0x00,
  0x03, 0x00, 0x00, 0x00, 0x3A, 0x00, 0x00, 0x00, 0x23, 0x00,
  0x00, 0x00, 0x51, 0x00, 0x00, 0x00, 0x46, 0x00, 0x00, 0x00,
  0x73, 0x00, 0x00, 0x00, 0x40, 0x00, 0x00, 0x00, 0x5E, 0x00,
  0x00, 0x00, 0x28, 0x00, 0x00, 0x00, 0x22, 0x00, 0x00, 0x00,
  0x06, 0x00, 0x00, 0x00, 0x18, 0x00, 0x00, 0x00, 0x4F, 0x00,
  0x00, 0x00
};
unsigned char c1[]={0x7d, 0x2d, 0x36, 0x37, 0x32, 0x49, 0x73, 0x6b,
0x47, 0x44, 0x17, 0xa, 0x3e, 0x44, 0x56,
 0x46, 0x48, 0x28, 0x24, 0x11, 0x52, 0x46, 0x41, 0x74, 0x52, 0x3,
0x3a, 0x23, 0x51, 0x46, 0x73, 0x40, 0x5e, 0x28, 0x22, 0x6, 0x18,
0x4f};

unsigned char key[]="1wesa234";
int main()
{
    unsigned char a;
    for(int i = 0; i < sizeof(c1);i+=1) {

        a = c1[i] ^ key[i % 8];
        printf("%c", a);
```

```
    }
}
```

Web

过滤了ph ht有关的后缀

危险函数是move_uploaded_file

利用1.php/.绕过

奶龙只检测文件内容的前5000，写垃圾数据绕过